

WEEK 3 • DAY 13

Domain Intelligence

Domains • WHOIS • Registrar • TLD

A structured study report on the Domain Name System (DNS): how human-readable domain names map to IP addresses, how domains are classified, and how WHOIS and registrars govern domain ownership and registration.



TOPIC REPORT

Prepared as part of Week 3 study track — Domain Intelligence

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01 Introduction to Domain Names

Every computer on the Internet is identified by a unique numeric address called an **IP address**, usually written in dotted-decimal notation (e.g. 172.16.122.204), which corresponds to a binary form (10101100 00010000 01111010 11001100). Machines use IP addresses to locate and communicate with one another, but such strings are difficult for people to remember.

The **Domain Name System (DNS)** solves this problem. It is a distributed database that translates human-friendly alphanumeric names — **domain names** — into the numeric IP addresses computers actually use. A domain name is simply a sequence of letters and/or numbers separated by one or more periods ("."), acting as a pointer to a unique location on the network.

An Everyday Analogy

Think of a **domain name** as a street address, and **DNS** as the address book of the Internet. Just as a courier looks up your address instead of your GPS coordinates, a browser looks up a domain name and DNS resolves it to the correct IP address behind the scenes.

In **www.google.com** or **www.yahoo.com**, the portion "yahoo.com" is the domain name, while the "www." prefix tells the browser to look for the World Wide Web interface of that domain. Domain names are far easier to recall than IP addresses — for instance, www.yahoo.com is much easier to remember than its underlying IP address, 69.147.76.15.

In short: a domain name is the string of letters associated with an IP address, and DNS is the mechanism that performs the conversion between the two.

02 Types of Domain Names & the TLD Hierarchy

DNS organizes all domain names in a **hierarchical structure**. At the top sit the **Top-Level Domains (TLDs)**, followed by second-level and third-level domains, and finally sub-domains.

2.1 Top-Level Domains (TLD)

The Top-Level Domain sits at the highest level of the DNS hierarchy and is sometimes called an "extension." TLDs fall into two broad categories:

Category	Description	Examples
Country Code TLD (ccTLD)	Two-letter domains, one for every country. Historically reserved mainly for residents of that country, though many ccTLDs now allow open registration. Used by companies to target a local audience.	.in (India), .au (Australia), .jp (Japan), .us (United States)
Generic TLD (gTLD)	Open for registration to anyone, regardless of citizenship, residence, or age.	.com (commercial), .net (network companies), .org (organizations), .edu (education), .biz (business)

2.2 Levels Below the TLD

Level	Position in Hierarchy	Example
Second-Level Domain	Directly below the TLD; also called the "label."	In .co.in → ".co" is the second-level domain under the ".in" ccTLD
Third-Level Domain	Directly below the second-level domain.	In yahoo.co.in → "yahoo" is the third-level domain under ".co", which sits under ".in"
Sub-domain	A named subdivision of a higher-level domain.	"yahoo.com" is a sub-domain of ".com"; "login.yahoo.com" is a sub-domain of "yahoo.com"

2.3 Advantages & Disadvantages of Domain Names

Advantages • Users do not need to remember numeric IP addresses. • More reliable and secure for everyday use than raw IPs.	Disadvantages • If an IP address changes (for several possible reasons), a DNS resolver may still have the previous IP cached, temporarily serving outdated or incorrect information.
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03 WHOIS — Looking Up Who Holds a Domain

WHOIS (pronounced “who is”) is a query-and-response protocol used to look up databases that store the registered users or assignees of an Internet resource — including domain names, IP address blocks, and autonomous systems — and to return that information in a human-readable format. The current version of the protocol is documented in RFC 3912. “Whois” is also the name of the command-line utility, available on most Unix-like systems, used to make these queries.

3.1 A Brief History

- The first WHOIS directory was created in the early 1970s at Stanford's Network Information Center (NIC) to look up information about people or entities on the ARPANET.
- Registration was originally handled by a single organization, so one centralized WHOIS server sufficed. As the commercial Internet grew, management passed through DARPA, then UUNET, and eventually to **InterNIC** (formed 1993) under a National Science Foundation contract.
- On 1 December 1999, management of the .com, .net, and .org TLDs was assigned to **ICANN**, which converted them to a “thin” WHOIS model — the model still broadly used today.
- The WHOIS protocol itself dates back to RFC 812 (1982), building on the earlier NAME/FINGER protocol (RFC 742, 1977).

3.2 Thick vs. Thin WHOIS Models

Model	How it Works	Trade-off
Thick WHOIS	One WHOIS server stores the complete registration data from all registrars for a given TLD (e.g. one server can answer for every .org domain).	Faster, more consistent lookups; if a registrar goes out of business, registration data is still retained centrally.
Thin WHOIS	The registry's WHOIS server stores only the name of the domain's registrar; the registrar's own server holds the full record (used by .com and .net).	If a registrar fails or data is lost, contact details may become hard to recover, risking the registrant's control of the domain.

3.3 Typical WHOIS Record Contents

A WHOIS record typically discloses the registry domain ID, the sponsoring registrar and its IANA ID, key dates (creation, last updated, expiry), domain status codes (e.g. clientTransferProhibited), the domain's authoritative name servers, and DNSSEC status. Where a registrant uses a **private/proxy registration** service, the registrar's contact details are shown in place of the registrant's own information.

3.4 Privacy, Misuse & the Move to RDAP

For most of its history, ICANN required domain WHOIS records to publicly expose a registrant's mailing address, phone number, and email. This openness helps law enforcement, journalists, and abuse-mitigation teams trace domain holders, but it has also been exploited by spammers and identity thieves harvesting contact data. The EU's **General Data Protection Regulation (GDPR)**, effective 25 May 2018, conflicted with this openness and forced ICANN to redact much personal data from public view.

This tension eventually led ICANN to replace WHOIS with the **Registration Data Access Protocol (RDAP)**, a more structured, access-controlled successor. Following an approved 2023 amendment to registry/registrar agreements, WHOIS was officially sunset for generic TLDs (gTLDs) on 28 January 2025,

with RDAP now the required service instead.

Quick Reference

- Default WHOIS server port: **TCP 43**
- Governing spec: **RFC 3912** (WHOIS Protocol Specification)
- Modern successor: **RDAP** (Registration Data Access Protocol)
- Related service: **RWhois** (Referral WHOIS) — a hierarchical, hardly-adopted extension defined in RFC 1714/2167, using TCP port 4321

04 Registrar — Who Manages Domain Registrations

A **domain name registrar** is a company, person, or office that manages the reservation of Internet domain names on behalf of registrants. To operate, a registrar must be accredited either by a generic top-level domain (gTLD) registry or by a country-code top-level domain (ccTLD) registry, and it must follow that registry's operating guidelines. As of March 2024, roughly 2,800 domain registrars were accredited by ICANN, with GoDaddy the largest, managing over 50 million .com domains.

4.1 How ICANN Accreditation Works

An **ICANN-accredited registrar** is an entity authorized to offer domain registration services for gTLDs with direct access to the gTLD registries. Any company wishing to sell domains under gTLDs such as .com, .net, or .org must first apply for and receive ICANN accreditation, meeting defined operational, technical, and financial standards. The relationship between ICANN and each accredited registrar is governed by a **Registrar Accreditation Agreement (RAA)**, and the relationship between the registrar and a registrant is governed by a separate registration agreement. Accredited registrars also pay ICANN a yearly accreditation fee. ICANN accreditation applies only to gTLDs — each ccTLD registry (for domains like .uk or .in) runs its own, independent accreditation process for registrars wishing to sell that country's domains.

4.2 Registry vs. Registrar vs. Registrant

Role	Function
Registry	Operates the authoritative database for an entire TLD (e.g. Verisign runs .com/.net) and maintains the master zone file.
Registrar	An ICANN- or ccTLD-accredited business (e.g. GoDaddy, Namecheap) that sells and manages individual domain registrations on the registry's behalf.
Registrant	The individual or organization that actually registers and owns the domain name.

4.3 A Parallel Concept: Registrar of Companies (ROC)

Outside the world of domains, the term “registrar” also appears in company law. In India, the **Registrar of Companies (ROC)** is a government authority under the Ministry of Corporate Affairs (MCA) responsible for registering and administering companies and Limited Liability Partnerships (LLPs) under the Companies Act, 2013. Its core functions — approving names, verifying documents, issuing a Certificate of Incorporation, maintaining an official register, and enforcing regulatory compliance — are conceptually similar to what a domain registrar does for domain names: both act as an authorized gatekeeper that verifies, records, and certifies who legitimately “owns” a name (be it a company name or a domain name) within a defined naming system.

Domain Registrar vs. ROC — the Analogy

- **Domain Registrar:** accredited by ICANN/ccTLD registries to register and manage domain names; issues WHOIS records for the domain.
- **ROC (India):** a government authority that registers companies/LLPs; issues a Certificate of Incorporation (COI) and a Corporate Identity Number (CIN).
- Both maintain a public register of who holds a given name/entity and both require ongoing compliance from the name-holder to keep the registration valid.

Summary

Together, **domain names**, the **TLD hierarchy**, **WHOIS**, and **registrars** form the backbone of how the Internet's naming system is organized, looked up, and governed: domain names make IP addresses human-readable; TLDs classify and hierarchically organize those names; registrars are the accredited businesses that sell and administer individual registrations; and WHOIS (now transitioning to RDAP) is the lookup layer that reveals who is responsible for a given domain.

Sources

GeeksforGeeks — "Introduction to Domain Name"; Wikipedia — "WHOIS"; Wikipedia — "Domain name registrar"; ICANN — "About Registrars" & Accredited Registrars pages; GeeksforGeeks — "Registrar of Companies (ROC): Full Form, Role, Functions and Scope".